

- Ortoogonalización

- Métodos:
1. Gram-Schmidt → Gram Schmidt → popularizó 1883, 1907
 2. GS modificado → 1820, Laplace
 3. Householder → 1958 (Oak Ridge National Lab)
 4. Givens → 1950 (Argonne National Lab)
 5. Re-ortogonalización

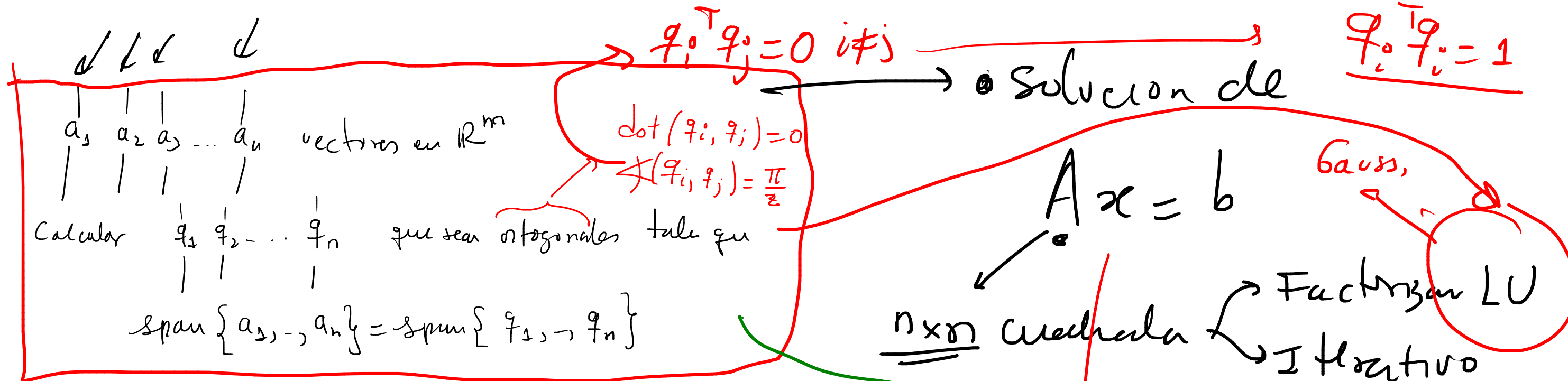
poco usado

menos
cálculos

mejor (estable)

más cálculos
& dispersa

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & x & x & 0 & 0 \\ 0 & x & x & 0 & 0 \\ 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & 0 & x \end{pmatrix}$$



$n < m$

$$\begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix}_{n \times m} \underbrace{\begin{bmatrix} | & & | \\ q_1 & q_2 & \dots & q_n \\ | & & | \end{bmatrix}}_{Q} = I_{n \times n}$$

$$m \times n$$

$n \times m$
 $n \neq q$
 $Qz = \tilde{b}$
 $Q^T Qz = Q^T \tilde{b}$
 $z \mapsto x = Qz$

$$A \rightarrow \begin{pmatrix} | & | & | & \dots & | \\ a_1 & a_2 & a_3 & \dots & a_n \\ | & | & | & \dots & | \\ \hline | & | & | & \dots & | \\ q_1 & q_2 & q_3 & \dots & q_n \\ | & | & | & \dots & | \end{pmatrix} \in \mathbb{R}^m \quad \underline{n < m}$$

$Q \rightarrow$

Triangular

$$A = QR$$

$$a_1 = r_{11} q_1$$

$$a_2 = r_{21} q_1 + r_{22} q_2$$

$$a_3 = r_{31} q_1 + r_{32} q_2$$

• Aplicaciones

① - Sistemas Lineales $m \times n$ $m \neq n$

Valores Singulares
- componentes principales } estadística
- Imágenes

② - Códigos

$$A = QR$$

Valores Propios.



- ① $A = QR$
- ② $A = RQ$
- ③ $A = QR$

$$A\varphi = \lambda\varphi$$

Valor Propio

$$\lambda \in \mathbb{C}$$

$$\varphi \in \mathbb{R}^n$$

$$\varphi \neq 0$$

vector propio

render
sistema
- sistemas
de E.D.O
- sistema E.D.P.

3. $Ax = b$

↓
muy grande y dispersa

$$\left(\frac{\text{nonzeros}(A)}{n^2} \right) \sim \begin{matrix} 5\% \\ 10\% \end{matrix}$$

- ~~QR~~, - QR iterativa

Métodos de Proyección $\left\{ \begin{array}{l} \rightarrow \\ \rightarrow \\ \rightarrow \end{array} \right\}$ Krylov: $\left\{ \begin{array}{l} C6 \\ 64 \text{ res} \end{array} \right\}$

1

$$Ax = b$$

minimos cuadrados
de curvas

para ajuste

$$n < m$$

ej:

x	y
x_1	y_1
x_2	y_2
$\vdots$	$\vdots$
x_m	y_m

$$y \approx a + b x + c x^2$$

$$y_1 \approx a + b x_1 + c x_1^2$$

$$y_2 \approx a + b x_2 + c x_2^2$$

$$y_m \approx a + b x_m + c x_m^2$$

$$\vec{\alpha} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 \end{bmatrix}$$

$$\vec{y} \approx A \vec{\alpha}$$

- $Q \in \mathbb{R}^{m \times p}$ $Q^T Q = I$ Q es ortogonal

$\rightarrow Q \in \mathbb{C}^{m \times p}$ $Q^* Q = I$ Q es unitaria

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_p \\ | & | & & | \end{bmatrix}$$

$$q_i^T q_j = \delta_{ij} = \begin{cases} 1 & \text{si } i=j \\ 0 & \text{si } i \neq j \end{cases}$$

$$Q Q^T = ?$$

si $m=p$
 $Q Q^T = I$

$$A = \begin{pmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_p \\ | & | & & | \end{pmatrix} \begin{pmatrix} r_{11} & r_{12} & \dots & 0 \\ 0 & r_{22} & \dots & 0 \\ 0 & 0 & r_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$m \times n$

$m \times p$

$p \times n$

Q

R

Factorización $A = QR$

$$Q^T Q = I_{p \times p}$$

$$\text{Imag}(A) = \text{span}(Q)$$

Factorization QR completa

$$A = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

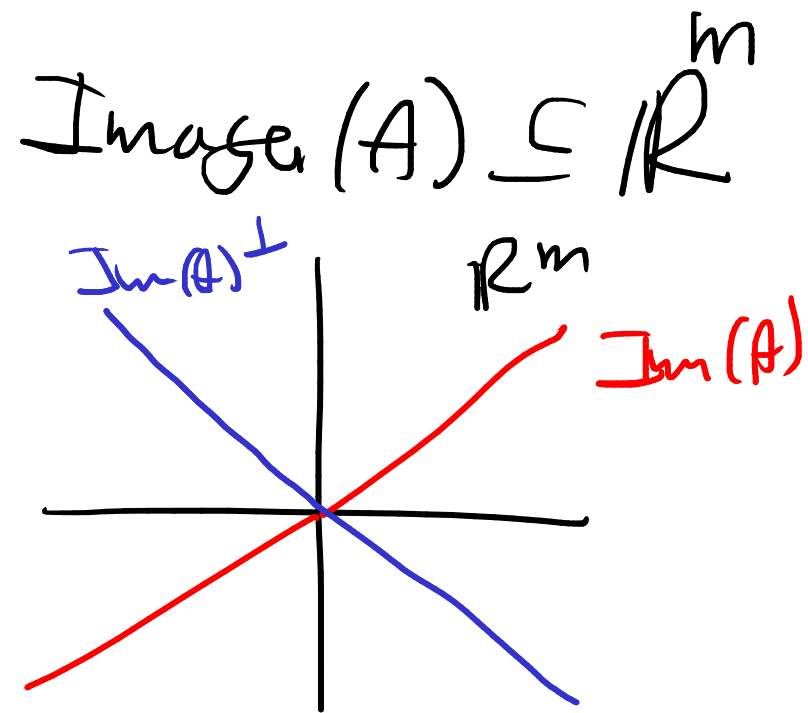
$m \times n$ $\begin{matrix} m \times p & m \times (m-p) \end{matrix}$ $m \times m$ $\begin{matrix} p \times n \\ (m-p) \times n \end{matrix}$ $m \times n$

$$Ax = b$$

$$\text{Image}(A) = \text{span } Q_1$$

$$\text{Image}(A)^\perp = \text{span } Q_2$$

Si las columnas de A son linealmente independientes $\boxed{p = n}$



Gram-Schmidt (clásico).

Dados $\{a_1, a_2, \dots, a_n\} \subseteq \mathbb{R}^m$ linealmente independientes

Generar $\{q_1, q_2, \dots, q_n\}$ tales que $q_i^T q_j = \delta_{ij}$ y

$$\begin{aligned} \text{span}\{a_1, a_2, \dots, a_n\} &= \text{span}\{q_1, a_2, a_3, \dots, a_n\} \\ &= \text{span}\{q_1, q_2, a_3, \dots, a_n\} \\ &\vdots \\ &= \text{span}\{q_1, q_2, \dots, q_{n-1}, a_n\} \\ &= \text{span}\{q_1, q_2, \dots, q_{n-1}, q_n\} \end{aligned}$$

$$\text{span}\{q_1\} = \text{span}\{a_1\} \quad q_1 = \pm \frac{a_1}{\|a_1\|}$$

$$\text{span}\{q_1, q_2\} = \text{span}\{a_1, a_2\}$$

$$\text{span}\{a_1, a_2, \dots, a_i\} = \text{span}\{q_1, \dots, q_i\}$$

$$\begin{aligned} a_2 &= r_{21} q_1 + r_{22} q_2 \\ q_2^T (r_{21} q_2 &= a_2 - r_{21} q_1) \rightarrow \underline{r_{22}} = q_2^T a_2 - 0 \\ q_1^T (r_{21} q_2 &= a_2 - r_{21} q_1) \rightarrow 0 = q_1^T a_2 - r_{21} \cdot 1 \end{aligned}$$

$$q_1, q_2, \dots, q_{i-1},$$

$$r_{ii} \quad q_i = a_i - \sum_{l=1}^{i-1} q_l r_{li}$$

$$\underline{f_j^T} \left(r_{ii} \quad f_i = a_i - \sum_{l=1}^{i-1} f_l \quad \textcircled{r_{li}} \right) \quad r_{ji}$$

$$0 = f_j^T a_i - r_{ji} \rightarrow \boxed{r_{ji} = f_j^T a_i}$$

$$f_i^T \left(r_{ii} \quad f_i = a_i - \sum_{l=1}^{i-1} f_l \quad \textcircled{r_{li}} \right) \rightarrow r_{ii} = f_i^T a_i.$$

Algorithm

Do $i = 1 : n$

Do $j = 1 : i - 1$

$$r_{ji} = q_j^T a_i$$

$$p_i = a_i - \sum_{k=1}^{i-1} q_k r_{ki}$$

$$r_{ii} = \|p_i\|$$

$$q_i = p_i / r_{ii}$$

End

QRCGS (generic function with 1 method)

```
• function QRCGS(B)
•     A=copy(B)
•     sizeA=size(A)
•     Q = zeros(sizeA)  #(m,n)
•     R = zeros(sizeA[2],sizeA[2])  #(n,n)
•     for i = 1:sizeA[2]
•         for j = 1:i-1
•             R[j,i] = Q[:,j]'A[:,i]
•         end
•         p = A[:,i] - Q[:,1:i-1]*R[1:i-1,i]
•         R[i,i]=norm(p)
•          # if abs(R[i,i])<0.000000000001
•          # println("Rii cercano 0")
•          # end
•         Q[:,i] = p/R[i,i]
•     end
•     return Q,R
• end
```